

Tajwar Razib

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Objective

I am a research-driven scholar with a solid background in Thermodynamics, Additive Manufacturing, Control System, and expertise in Machine Learning. My work combines computational and data-driven approaches to investigate energy systems and manufacturing process control. In an effort to broaden my scope of research, I am committed to interdisciplinary exploration and academic advancement in various scientific domains.

Education

Graduate Teaching Assistant
Rensselaer Polytechnic Institute

August 2026- Present

Bachelor of Science (BSc) in Mechanical Engineering
Bangladesh University of Engineering and Technology (BUET)
CGPA: 3.68 out of 4.00

June 2026

Technical Skills

- **Programming Language:** Python, C, C++
- **Software:** MATLAB, AutoCAD, SolidWorks, COMSOL, LAMMPS
- **Plot Analysis Tools:** OriginPro, Tecplot, xyExtract, Polymath Professionals

Academic Projects

Project 1 Saylobot: Digital Data Acquisition System for Saybolt Viscometer

Here, my team automated the reading taking from the Saybolt Viscometer with the necessary sensors and Arduino program. I was the CAD designer for the project.

Project 2 Shell and Tube heat exchanger with inclined baffles

My team and I built a Shell and Tube heat exchanger with an inclined baffle. I was the Python programmer for heat exchange analysis.

Undergraduate Thesis

Machine Learning based Power Prediction and Organic Rankine Cycle Integration for Waste Heat Recovery in a Combined Cycle Power Plant

Publications

Journal Publications

- **T. Razib, A. Saha.** *A comprehensive thermodynamic analysis of a bottoming organic Rankine cycle using optimized working fluids* <https://doi.org/10.1016/j.nxener.2026.100654>

Preprints & Under Review

- A. Sarker, **T. Razib**, A. Saha., S. Saha *LLM Agent-guided Stochastic Control of Multi-layer and Multi-track Laser Powder Bed Fusion Process*. 10.21203/rs.3.rs-9315895/v1

Conference Proceedings

- **T. Razib**, A. Saha. *Thermodynamic analysis of a bottoming organic rankine cycle for waste heat recovery* <https://doi.org/10.1063/5.0327926>
- A. Nafis, A.M. Momtaz, **T. Razib**, A. Saha. *A Comprehensive Study on Energy, COP, and Exergy of a Coupled ORC-VCC Cogeneration System Employing Dual Working Fluids*. <https://dx.doi.org/10.2139/ssrn.6204778>
- **T. Razib**, S. A. Tapu *Application of Improved Particle Swarm Optimization (PSO) on a Gas Turbine Model*. <https://doi.org/10.46254/BA08.2025>

Manuscripts in Preparation

1. Physics-Informed Disentanglement of Multimodal Data on Additive Manufacturing by Variational Auto-Encoder.
2. Comparative Analysis of PI and PID Controllers for Traction Motors in Hybrid Electrical Vehicles Using Multi-Objective Optimization via NSGA-III.

Abstract Accepted

1. Application of Bayesian Optimization on Design and Working Parameters of an Inclined Baffle Shell and Tube Heat Exchanger. *ASTFE 2026*

References

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